ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 2 [10 marks]

Q2(a) [2 marks]

$\dot{x}_1 = x_2$ and $\dot{x}_2 = (\tau - b\,x_2 + mg\ell \sin\theta)/J$. At equilibrium:

$$_0 = -mg\ell sin\theta_e,$$

an equilibrium exists for $|\tau_0|$ less than or equal to 2.4525 N m

Q2(b) [2 marks]

The Jacobian has $A_{12} = 1$ and $A_{22} = -b/J = -2.5$. The input matrix is $B = [0, 50]^T$.

Q2(c) [4 marks]

At steady state the derivative terms vanish. The proportional torque is $K_p(\theta_d - \theta)$, which must balance $mg\ell \sin\theta$.

Q2(d) [2 marks]

Gravity compensation makes the steady-state error zero. It does not change the closed-loop poles because it only adds a constant torque.

Question 3 [10 marks]

Q3(a) [2 marks]

Q3(b) [2 marks]

Q3(c) [4 marks]

A-LC=[[$-\ell_1$,1], [$-\ell_2$,-2]]. I obtain $p(s)=s^2+(\ell_1+2)s+(\ell_1+\ell_2)$. Matching $(s+10)^2$ gives $\ell_1=18$ and $\ell_2=82$.

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

Question 4 [10 marks]

Q4(a) [2 marks]

The characteristic polynomial is s^2+4s+8, so I report poles -4 and -8. Unity DC gain requires Nff=8.

Q4(b) [2 marks]

The nominal controller has unity DC gain, so θss=1 and the tracking error is zero even after the actuator gain changes.

Q4(c) [4 marks]

$$k_1 = 18, k_2 = 6, K_I = 32$$

Q4(d) [2 marks]

Integral action gives zero steady-state error if stable. A practical problem is integrator windup.